

Health Advisory App

Transforming Data Into Better Health Choices



Meet the Team:

Mohitha G

Ridhi Verma

Sushma Chowdary M

TECH STACK USED:

01 Operating Environment

- Windows Subsystem for Linux (WSL)

02 Backend Components

- Pathway
- LLM-App Repository
- Retrieval-Augmented Generation (RAG) Pipelines
- Langchain

03 Frontend Components

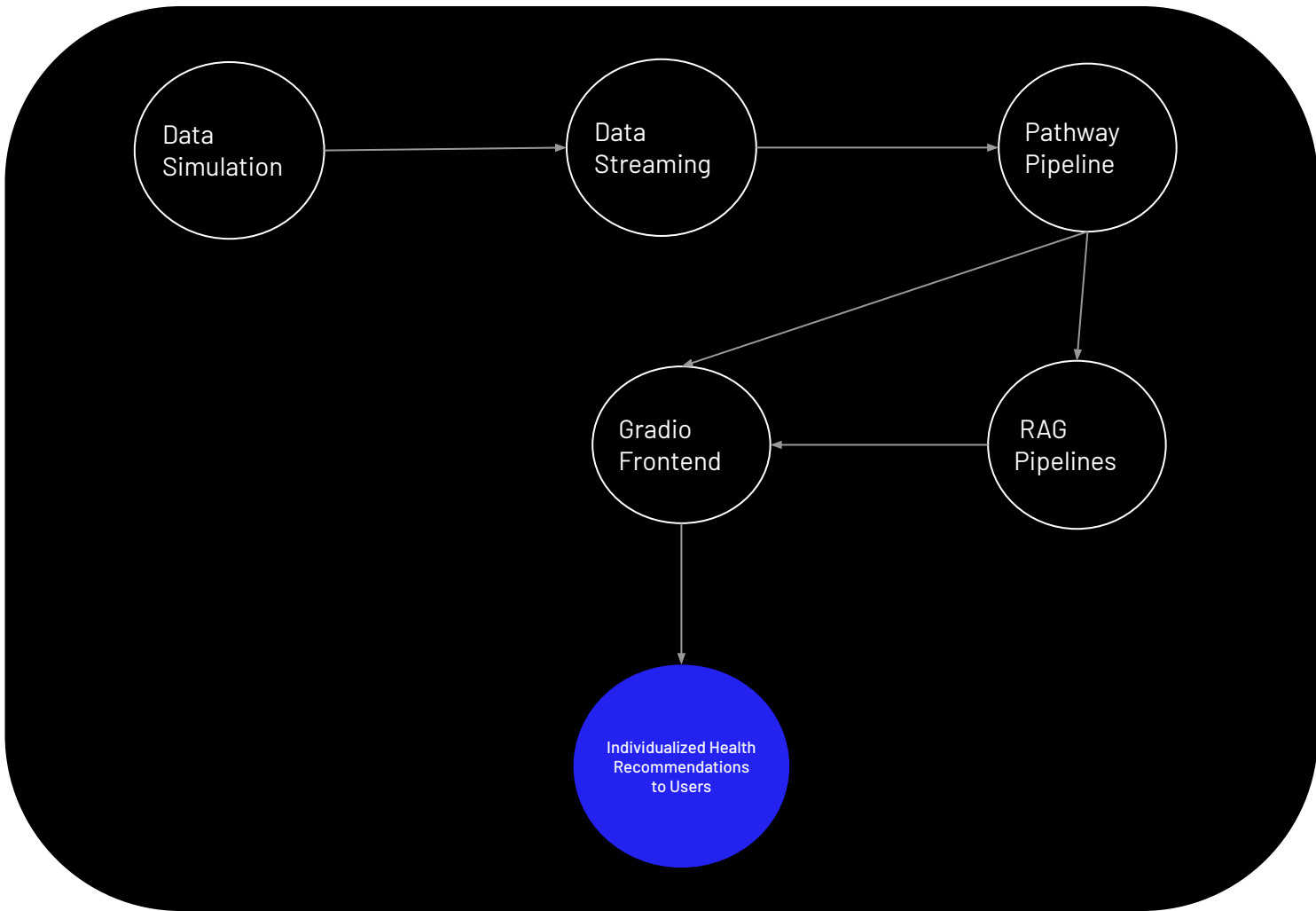
- Gradio

04 Language : Python

01

Architecture

Architecture diagram



STEP TO STEP APPROACH

Data Simulation

- Randomly generates health metrics like timestamp, heart rate, steps, calories, and sleep hours using `simulate_data.py`. This data is saved to a CSV file (`health_data.csv`).

Data Streaming

- Streams real-time data from the CSV file to enable continuous updates using `stream_data.py`.

Pathway Pipeline

- Processes incoming data streams, ensuring seamless integration with Gradio and RAG pipelines using `pathway_pipeline.py`. Pathway handles data transformation and real-time insights generation.

STEP TO STEP APPROACH

Gradio Frontend

- `Health_advisory_app.py` provides an interface where all can view their health metrics and receive recommendations in real time using gradio.

RAG Pipelines (Retrieval-Augmented Generation)

- Pathway and the provided LLM-App repository are integrated so that it can provide the personalised recommendations by the real time data using RAG.

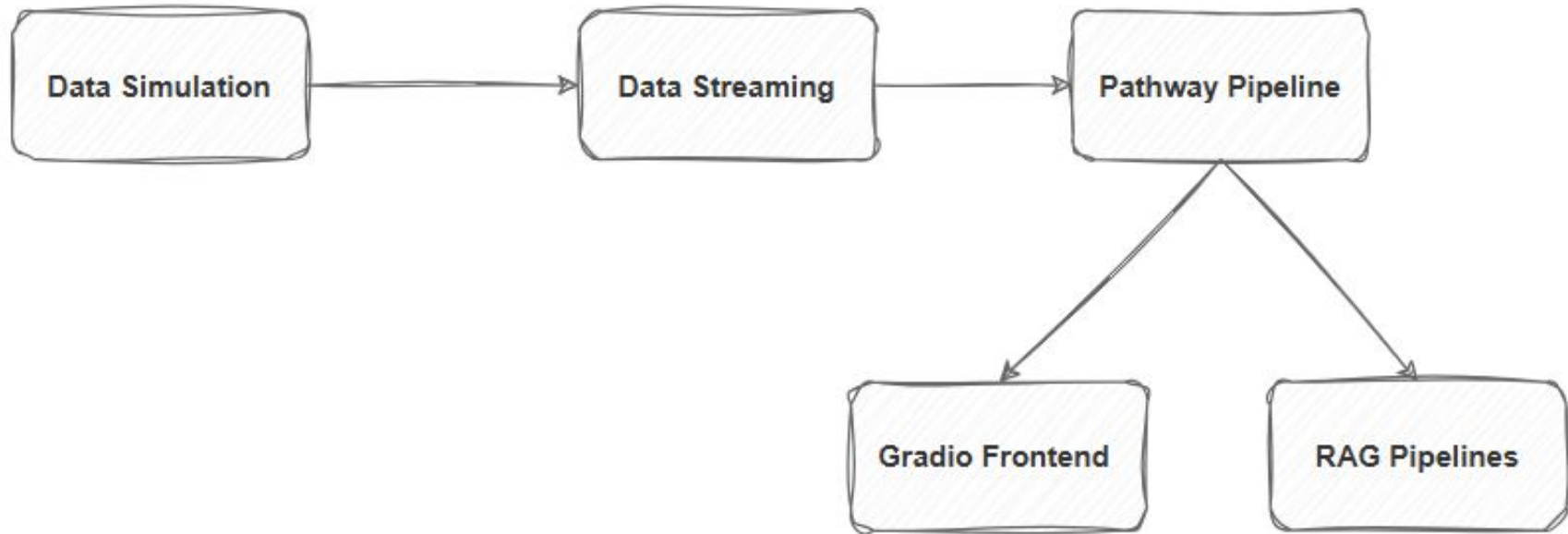
Final Output

- **Individualized Health Recommendations:**
 - Users receive actionable advice tailored to their real-time health data

02

Data Flow

Data Flow throughout the application



DATA FLOW EXPLANATION

DATA SIMULATION

Generates random health metrics – heart rate, steps, calories, sleep hours.

DATA STREAMING

Streams the simulated data in real-time.

PATHWAY PIPELINE

Processes the data for real-time insights and connects to the next components.

GRADIO FRONT-END & RAG PIPELINES

The data acts as knowledge sources that enhances the insights which help RAG to give a suitable recommendation to the gradio frontend that will be displayed to user

03

RAG pipelines insights

Insights

Integrating Retrieval-Augmented Generation (RAG) with Pathway enables real-time, personalized health advisories by combining dynamic data processing with contextual intelligence. Pathway ingests and processes streaming data from wearables, detecting anomalies or conditions (e.g., elevated heart rate or low steps). These insights are sent as queries to the RAG pipeline, which retrieves domain-specific advice from a VectorStore and uses a language model to generate tailored recommendations (e.g., "Your heart rate is high; stay hydrated and take a short walk."). The enriched data stream is then displayed via a Gradio interface, providing dynamic, user-friendly insights. This integration ensures scalability, personalization, and adaptability for real-time health monitoring systems.

04

Real-World Impact

01

Improved Preventive Healthcare

By analyzing real-time health data such as heart rate, steps, and sleep patterns, the system can proactively identify potential health issues before they become critical

02

Personalized Health Recommendations

These personalized suggestions are designed to enhance well-being and promote healthier lifestyle choices. By analyzing individual data, the system identifies key factors that contribute to overall health improvements.

03

Enhanced Decision-Making

These personalized suggestions are designed to enhance well-being and promote healthier lifestyle choices. By analyzing individual data, the system identifies key factors that contribute to overall health improvements.

THANK YOU